



Data and Artificial Intelligence

Cyber Shujaa Program

Week 11 Project Report NLP Using Transformers
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1.0 INTRODUCTION

This project explores the application of Natural Language Processing (NLP) using BERT from Hugging Face Transformers. The primary objective was to evaluate semantic similarity between sentence pairs using BERT's contextual embeddings.

Unlike traditional approaches like Bag-of-Words or TF-IDF, BERT provides context-aware embeddings for words, which makes it ideal for distinguishing meaning based on context (e.g., "bank" of a river vs. "bank" as a financial institution).

2.0 TASK COMPLETION

2.1 Loading and Preparing the Dataset

The dataset for this project was manually defined as a list of 15 sentence pairs, including both semantically similar and dissimilar examples. These included questions and phrases related to programming, food, computers, nature, and polysemous words like “bank” and “crane”.

The dataset was encoded using the BertTokenizer and embeddings were extracted from the [CLS] token using the TFBertModel.

```
1 sentence_pairs = [  
2     ("How do I learn Python?", "What is the best way to study Python?"),  
3     ("What is AI?", "How to cook pasta?"),  
4     ("How do I bake a chocolate cake?", "Give me a chocolate cake recipe."),  
5     ("How can I improve my coding skills?", "Tips for becoming better at programming."),  
6     ("Where can I buy cheap laptops?", "Best sites to find affordable computers."),  
7     ("I love eating mangoes in summer.", "Mangoes taste best during hot seasons."),  
8     ("The bank will close at 5 PM today.", "He sat by the river bank and read."),  
9     ("She made a quick run to the store.", "She sprinted fast to buy groceries."),  
10    ("The crane lifted the heavy beam.", "I saw a crane flying over the lake."),  
11    ("He scored a goal in the final match.", "She submitted her final thesis."),  
12    ("The sun sets over the horizon.", "The moon rises from the sky."),  
13    ("He is planning a trip to the mountains.", "She is preparing a dinner meal."),  
14    ("I'm waiting for the bus to arrive.", "I'm waiting for the train to arrive."),  
15    ("She is making a cup of tea.", "He is making a cake."),  
16    ("She is a mother to two young children.", "He is a father to a young child.")  
17 ]  
18  
19 # Ground Truth Labels: 1 = Similar, 0 = Not Similar  
20 # labels = [1, 0, 1, 1, 1, 1, 0, 1, 0, 0]  
21 labels = [1, 0, 1, 1, 1, 1, 0, 1, 0, 0, 1, 1, 1, 1, 1]
```

✓ 0.0s

```

5 # Prepare DataFrame
6 df = pd.DataFrame({
7     "Pair ID": list(range(1, len(sentence_pairs) + 1)),
8     "Sentence 1": [s[0] for s in sentence_pairs],
9     "Sentence 2": [s[1] for s in sentence_pairs],
10    "Cosine Similarity": scores,
11    "Predicted": predictions,
12    "Ground Truth": labels
13 })
14
15 df["Sentence Pair"] = df["Sentence 1"] + " ↔ " + df["Sentence 2"]
16

```

2.2 Data Preprocessing

Preprocessing involved:

- Tokenization using `BertTokenizer.from_pretrained('bert-base-uncased')`
- Padding and truncation
- Extraction of embeddings from the [CLS] token

A class-based design was implemented using Python's **OOP paradigm** to wrap these steps in a reusable `BERTSimilarityEngine` class.

Each sentence was passed through the model, and cosine similarity was computed between the [CLS] embeddings of each pair.

```

3 def __init__(self, threshold=0.7):
4     # Load tokenizer and model
5     self.tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')
6     self.model = TFBertModel.from_pretrained('bert-base-uncased')
7     self.threshold = threshold
8
9     Tabnine | Edit | Test | Explain | Document
10    def get_cls_embedding(self, sentence):
11        # Tokenize and encode
12        inputs = self.tokenizer(sentence, return_tensors='tf', padding=True, truncation=True)
13        outputs = self.model(inputs)
14        # Extract the [CLS] token (first position)
15        cls_embedding = outputs.last_hidden_state[:, 0, :]
16        return cls_embedding.numpy()
17

```

```
1 engine.get_cls_embedding(sentence_pairs[0][0])
[6] ✓ 1.0s
... array([[ -3.91852036e-02,  2.00920671e-01,  1.42418295e-02,
          -4.57232669e-02, -2.91943014e-01,  1.55994594e-02,
           7.30915546e-01,  4.47535932e-01, -2.41496414e-01,
           1.58486843e-01, -7.40697011e-02, -1.75546169e-01,
          -1.09237850e-01, -1.04426518e-02,  3.90605181e-01,
          -8.40551257e-02, -1.34050623e-01,  3.91230404e-01,
           2.96944469e-01,  2.25489885e-01,  8.91168416e-02,
           2.85556726e-03, -1.59137815e-01, -2.80068778e-02,
           1.36740595e-01, -9.25886706e-02,  1.31215885e-01,
           9.62996036e-02,  2.41178200e-02, -1.74419105e-01,
           1.36259198e-01,  7.11333901e-02, -1.05665490e-01,
          -2.79373854e-01,  1.78407237e-01, -1.54735342e-01,
          -9.88034979e-02, -1.42643452e-01,  8.13576728e-02,
           7.21337497e-02, -3.76233995e-01,  1.09871104e-01,
          -2.92348564e-01,  1.76500693e-01,  1.66223750e-01,
          -4.67850655e-01, -2.22842360e+00, -4.50845718e-01,
          -4.83272284e-01, -1.96760133e-01, -3.71105336e-02,
           2.72407323e-01,  4.07993555e-01,  2.26447582e-01,
          -2.19611242e-01,  1.55035958e-01, -9.85073596e-02,
           4.01863694e-01, -2.25866541e-01,  1.95919685e-02,
           2.32647061e-01,  1.47371739e-01, -2.01571390e-01,
           2.41106823e-02, -2.40531936e-01,  3.37097585e-01,
           1.17358938e-03,  2.79671162e-01, -1.07472584e-01,
           3.48274410e-01, -1.39723331e-01,  3.83727252e-02,
           9.68086123e-02,  1.91130698e-01, -2.46867463e-01,
          ...
          -1.79201603e-01, -1.72303617e-03, -5.33728123e-01,
           2.32454315e-01, -4.99318957e-01,  7.47929066e-02,
          -1.04025304e-01,  3.46019268e-02, -2.68947780e-01,
          -1.33368805e-01,  2.38330126e-01, -2.07461968e-01,
          -6.66839108e-02,  3.48497808e-01,  7.05615878e-01]], dtype=float32)
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```

```
1 engine.get_cls_embedding("She made a quick run to the store.")
[7] ✓ 0.3s
... array([[ 1.94903597e-01, -8.26063529e-02, -1.97548747e-01,
           1.31812170e-02,  1.44846231e-01, -3.12832922e-01,
           2.35954672e-03,  6.52519763e-01,  2.61001348e-01,
          -2.40377009e-01,  1.18824050e-01, -1.41004890e-01,
           5.80372922e-02, -1.72182962e-01,  2.11199209e-01,
          -7.39363432e-02, -2.93122649e-01, -8.09094831e-02,
          -7.20383599e-03, -2.65150145e-02, -9.18813348e-02,
          -3.73964906e-01,  1.79698765e-01,  5.36940038e-01,
           2.98572063e-01, -3.64454314e-02, -1.88356414e-01,
           2.11771838e-02,  3.29418331e-01,  1.14464566e-01,
          -1.97535262e-01,  2.68949509e-01, -4.14146692e-01,
          -2.58391112e-01,  1.45895422e-01, -1.34626463e-01,
           3.75526279e-01, -8.25229660e-02,  3.13938186e-02,
           2.91352630e-01, -8.17624807e-01, -3.48675579e-01,
          -5.37271239e-02,  2.24469811e-01,  4.03735995e-01,
          -1.83318853e-01, -3.18210530e+00, -1.77990198e-01,
          -9.38765332e-02, -3.92639637e-01,  2.63747185e-01,
          -3.33689153e-01,  1.99058890e-01,  4.14670557e-01,
           2.55792141e-01,  4.89953369e-01, -4.00132239e-01,
           4.05386508e-01,  3.77540253e-02,  2.45814510e-02,
           8.08316231e-01,  1.08909078e-01, -5.95993958e-02,
           2.26258367e-01, -2.27785073e-02, -2.61088073e-01,
           1.10685170e-01, -1.72079489e-01,  2.90766358e-02,
           7.02830017e-01, -5.91324627e-01,  7.56615698e-02,
           3.53595197e-01, -1.94178283e-01, -9.07288194e-02,
          ...
          -2.26002991e-01,  2.25885421e-01, -1.28075495e-01,
           7.40904361e-02, -3.44425112e-01, -2.67103493e-01,
          -3.39480713e-02,  4.58459109e-01, -2.11905926e-01,
          -1.38147958e-02, -1.17502347e-01, -4.56039637e-01,
           1.21854700e-01,  9.38347504e-02,  6.49599552e-01]], dtype=float32)
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```

```

1 # Define the BERT Similarity class
2 class BERTSimilarityEngine:
3     def __init__(self, threshold=0.7):
4         # Load tokenizer and model
5         self.tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')
6         self.model = TFBertModel.from_pretrained('bert-base-uncased')
7         self.threshold = threshold
8
9     def get_cls_embedding(self, sentence):
10        # Tokenize and encode
11        inputs = self.tokenizer(sentence, return_tensors='tf', padding=True, truncation=True)
12        outputs = self.model(inputs)
13        # Extract the [CLS] token (first position)
14        cls_embedding = outputs.last_hidden_state[:, 0, :]
15        return cls_embedding.numpy()
16
17    def compute_similarity(self, sent1, sent2):
18        # Get embeddings
19        emb1 = self.get_cls_embedding(sent1)
20        emb2 = self.get_cls_embedding(sent2)
21        # Compute cosine similarity
22        score = cosine_similarity(emb1, emb2)[0][0]
23        # Predict label
24        label = 1 if score > self.threshold else 0
25        return score, label
26
27    def evaluate(self, pairs, labels):
28        predictions = []
29        scores = []
30
31        print("\n🔍 Sentence Similarity Evaluation:\n")
32        for i, (s1, s2) in enumerate(pairs):
33            score, pred = self.compute_similarity(s1, s2)
34            scores.append(score)
35            predictions.append(pred)
36            print(f"Pair {i+1}")
37            print(f"Sentence 1: {s1}")
38            print(f"Sentence 2: {s2}")
39            print(f"Cosine Similarity: {score:.4f} → Predicted: {'Similar' if pred else 'Not Similar'}\n")
40
41        correct = sum([1 for p, l in zip(predictions, labels) if p == l])
42        accuracy = correct / len(labels)
43        print(f"Final Accuracy: {accuracy:.2%}")
44        return predictions, accuracy, scores

```

[2] ✓ 0.0s

2.3 Model Implementation and Evaluation

The model compared all 15 sentence pairs using cosine similarity. A threshold of **0.7** was used to classify similarity:

- If similarity > 0.7 - **Label: 1 (Similar)**
- Else - **Label: 0 (Not Similar)**

```

17    def compute_similarity(self, sent1, sent2):
18        # Get embeddings
19        emb1 = self.get_cls_embedding(sent1)
20        emb2 = self.get_cls_embedding(sent2)
21        # Compute cosine similarity
22        score = cosine_similarity(emb1, emb2)[0][0]
23        # Predict label
24        label = 1 if score > self.threshold else 0
25        return score, label
26
27    def evaluate(self, pairs, labels):
28        predictions = []
29        scores = []
30
31        print("\n🔍 Sentence Similarity Evaluation:\n")
32        for i, (s1, s2) in enumerate(pairs):
33            score, pred = self.compute_similarity(s1, s2)
34            scores.append(score)
35            predictions.append(pred)
36            print(f"Pair {i+1}")
37            print(f"Sentence 1: {s1}")
38            print(f"Sentence 2: {s2}")
39            print(f"Cosine Similarity: {score:.4f} → Predicted: {'Similar' if pred else 'Not Similar'}\n")
40
41        correct = sum([1 for p, l in zip(predictions, labels) if p == l])
42        accuracy = correct / len(labels)
43        print(f"Final Accuracy: {accuracy:.2%}")
44        return predictions, accuracy, scores

```

✓ 0.0s

Final Accuracy: 73.33%

Sentence Similarity and Evaluation

```
🔍 Sentence Similarity Evaluation:

Pair 1
Sentence 1: How do I learn Python?
Sentence 2: What is the best way to study Python?
Cosine Similarity: 0.9743 → Predicted: Similar

Pair 2
Sentence 1: What is AI?
Sentence 2: How to cook pasta?
Cosine Similarity: 0.9033 → Predicted: Similar

Pair 3
Sentence 1: How do I bake a chocolate cake?
Sentence 2: Give me a chocolate cake recipe.
Cosine Similarity: 0.8938 → Predicted: Similar

Pair 4
Sentence 1: How can I improve my coding skills?
Sentence 2: Tips for becoming better at programming.
Cosine Similarity: 0.8633 → Predicted: Similar

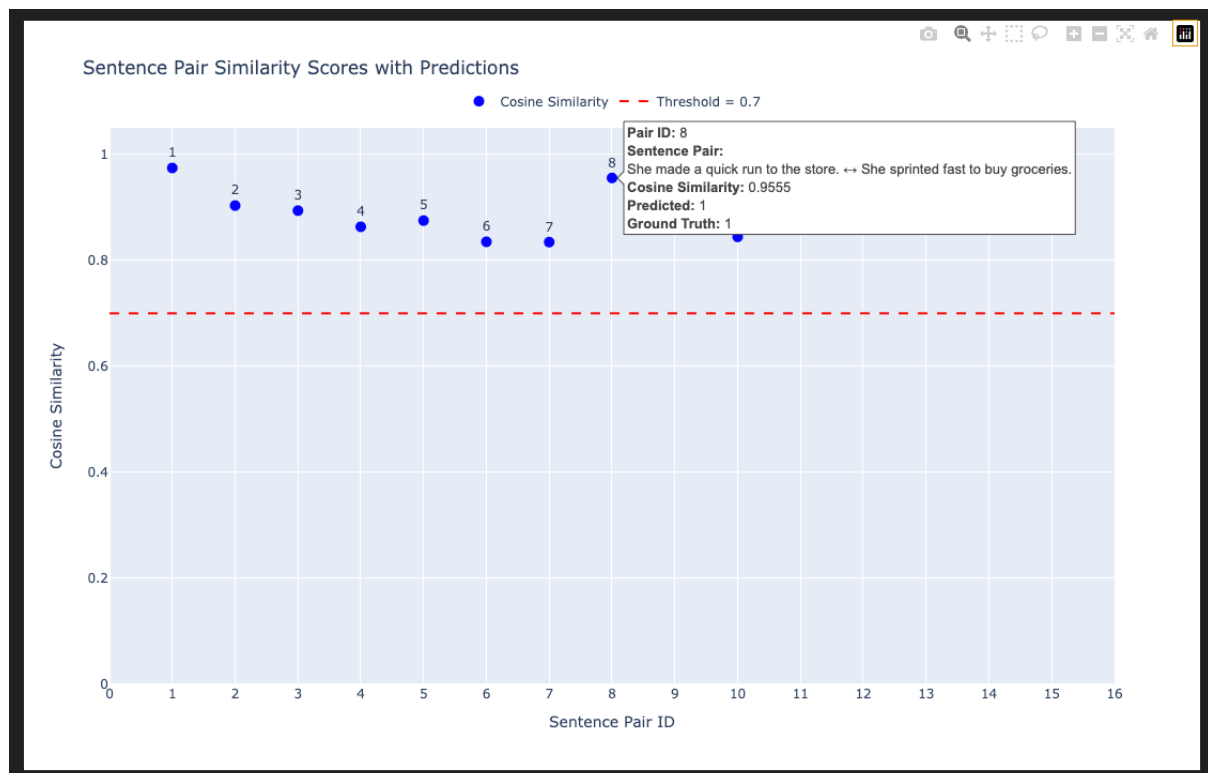
Pair 5
Sentence 1: Where can I buy cheap laptops?
...
Sentence 2: He is a father to a young child.
Cosine Similarity: 0.9099 → Predicted: Similar

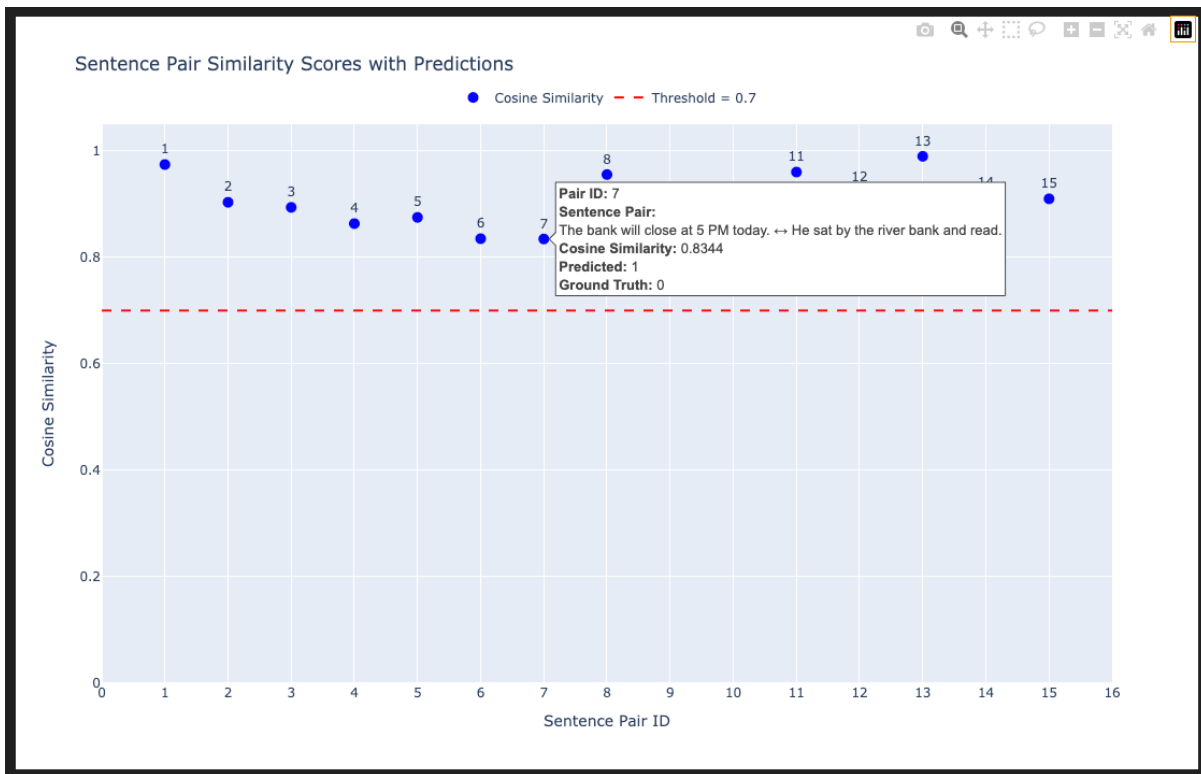
Final Accuracy: 73.33%
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```

Ground truth labels were manually assigned. The model predicted similarity based on computed embeddings and achieved a **final accuracy of 73.33%**.

2.4 Model Visualization

A line plot was generated to visualize cosine similarity scores across all sentence pairs.





- X-axis: Sentence Pair ID (1 to 15)
- Y-axis: Cosine Similarity Scores
- Horizontal line at 0.7 shows the threshold

The line chart demonstrates how BERT embeddings vary across sentence pairs, with the 0.7 threshold line visually separating similar vs non-similar predictions. Outliers and borderline cases are clearly distinguishable.

2.5 Publishing and Sharing of Notebook

The final notebook was developed and tested in VS Code, and uploaded to Google Drive with public accessibility.

 Notebook Link:

[<https://drive.google.com/drive/folders/1nGfdU8TnFSneX50UnxId-xAIHQU9wvFB?usp=sharing>]

3.0 PORTFOLIO WEBSITE

 Website: <https://amidu-dabor.github.io/>

This project has also been documented in the "Projects" section of the above portfolio.

4.0 CONCLUSION

This project significantly enhanced my understanding of transformer-based models, particularly BERT, and demonstrated their superiority over traditional NLP approaches such as Bag-of-Words and TF-IDF. By engaging in hands-on implementation, I explored how BERT captures contextual semantics at the sentence level, enabling more nuanced and accurate comparisons between sentence pairs.

One of the most impactful realizations was how BERT's [CLS] token can effectively represent an entire sentence's meaning by considering the context of all surrounding words. This context-aware representation allowed for precise measurement of similarity using cosine distance, and applying a fixed threshold of 0.7 proved to be an effective strategy for binary classification of sentence similarity.

The project also emphasized the value of clean, reusable code through object-oriented programming. Encapsulating the core functionality within a `BERTSimilarityEngine` class not only improved modularity and readability but also made it easier to extend or modify the code later.

Finally, visualizing the similarity scores brought an important layer of interpretability to the process. The ability to observe how sentence pairs relate in semantic space, with a clear visual threshold, made the model's behaviour more transparent and explained its decisions beyond raw numerical output.

Overall, this project bridged theoretical concepts with practical execution, reinforcing my skills in both modern NLP techniques and professional coding practices.

5.0 REFERENCES

1. Hugging Face Transformers
<https://huggingface.co/docs/transformers>
2. TensorFlow Documentation
<https://www.tensorflow.org/>
3. Scikit-learn Cosine Similarity
https://scikit-learn.org/stable/modules/generated/sklearn.metrics.pairwise.cosine_similarity.html

4. Python Official Docs

<https://docs.python.org/3/>

5. Matplotlib Documentation

<https://matplotlib.org/stable/contents.html>